Georgios Pantazopoulos

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Education

PhD in Robotics and Autonomous Systems **Sep 2021 - present**
CDT in Robotics and Autonomous Systems, Heriot-Watt University, University of Edinburgh

- Thesis: Inductive Biases in Architectures of Modern Visual Language Models, supervised by A. Suglia

MSc in Artificial Intelligence with Speech and Multimodal Interaction **Sep 2020 - Aug 2021**
Heriot-Watt University

- Grade: 84/100 (1st with Distinction)
- Thesis: Compositional Generalization in Image Captioning, supervised by A. Eshghi

MEng in Electrical and Computer Engineering **Sep 2012 - Jun 2018**
National Technical University of Athens

- Grade: 7.7/10, Major: Computer Science
- Thesis: Stress Detection from Speech Signals, supervised by A. Potamianos

Publications

An Efficient Training Pipeline for Reasoning Graphical User Interface Agents
2026 ICLR 2026 Workshop on Multimodal Intelligence, [paper]

- **G.Pantazopoulos**, E.Özyiğit

Evaluating Multimodal Language Models as Visual Assistants for Visually Impaired Users
2025 Annual Meeting of the Association for Computational Linguistics, [paper]

- A.Karamolegkou*, M.Nikandrou*, **G.Pantazopoulos***, D.Sanchez, P.Rust, R.Dhar, D.Hershcovich, A.Søgaard

CROPE: Evaluating In-Context Adaptation of Vision and Language Models to Culture-Specific Concepts
2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics, [paper]

- M. Nikandrou, **G. Pantazopoulos**, N. Vitsakis, I. Konstas, A. Suglia

Shaking Up VLMs: Comparing Transformers and Structured State Space Models for Vision & Language Modeling
2024 Conference on Empirical Methods in Natural Language Processing, [paper]

- **G. Pantazopoulos**, M.Nikandrou, A. Suglia, O. Lemon, A. Eshghi

Enhancing Continual Learning in Visual Question Answering with Modality-Aware Feature Distillation
2024 ACL Third Workshop on Advances in Language and Vision Research, [paper]

- M.Nikandrou*, **G. Pantazopoulos***, I. Konstas, A. Suglia

Lost in Space: Probing Fine-grained Spatial Understanding in Vision and Language Resamplers
2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics, [paper]

- **G. Pantazopoulos**, A. Suglia, O. Lemon, A. Eshghi

Learning To See But Forgetting To Follow: Visual Instruction Tuning Makes LLMs More Prone To Jailbreak Attacks
2024 LREC-COLING Third Workshop on Safety for Conversational AI, [paper]

- **G. Pantazopoulos**, A. Parekh, Malvina Nikandrou, A. Suglia

Multitask Multimodal Prompted Training for Interactive Embodied Task Completion
2023 Conference on Empirical Methods in Natural Language Processing, [paper]

- **G. Pantazopoulos**, M.Nikandrou, A. Parekh, B. Hemanthage, A. Eshghi, I. Konstas, V. Rieser, O. Lemon, A. Suglia

EMMA: A foundation model for embodied, interactive, multimodal task completion in 3D environments
2023 Alexa Prize SimBot Challenge Proceedings, [paper]

- A. Parekh*, M.Nikandrou*, **G. Pantazopoulos***, B. Hemanthage, A. Eshghi, I. Konstas, O. Lemon, A. Suglia

Combine to Describe: Evaluating Compositional Generalization in Image Captioning
2022 ACL Student Research Workshop, [paper]

- **G. Pantazopoulos**, A. Suglia, A. Eshghi

Data Augmentation using GANs for Speech Emotion Recognition
2019 Annual Conference of the International Speech Communication Association, [paper]

- A. Chatziagapi, G. Paraskevopoulos, D. Sgouropoulos, **G. Pantazopoulos**, M.Nikandrou, T. Giannakopoulos, A. Katsamanis, A. Potamianos, S. Narayanan

Projects

Retrievit **Dec 2025 - Mar 2026**
University of Edinburgh, [*code*]

- Benchmarking Transformers, State Space Models, and Hybrid models on long sequence modeling

EMMA **Jan 2022 - May 2023**
Alexa Prize Simbot Challenge, Finalist [*code*]

- Developed an interactive language-guided agent executing instructions in simulated environments

Exploring the Role of Patch Resolution for Vision Transformers **Jan 2022 - Mar 2022**
University of Edinburgh

- Explored the trade-off between performance gains and inference cost on downstream tasks

Industry Experience

AI Research Intern **Jun 2025 - Dec 2025**
Alan Turing Institute

- Worked on reasoning models for GUI agents
- Worked on data pipelines for filtering noisy data for training GUI agents

AI Engineer Intern **Feb 2025 - Apr 2025**
Reka AI

- Implemented contrastive methods with hard-negative mining for image-to-image, image-to-video, image & text-to video retrieval
- Contributed to real-time object detection and instance retrieval

Machine Learning Engineer **Sep 2018 - Apr 2020**
Behavioral Signal Technologies Inc.

- Developed CNN-based classification models for emotion recognition from raw audio signals
- Built LSTM-based classification models for emotion recognition from automatic transcriptions
- Implemented late and early-fusion methods for audio and text inputs improving by 8% over unimodal models
- Lead the data collection from crowdsourcing platforms resulting in the collection of 600k annotated examples
- Implemented active learning methods improving the performance in minority classes by 10%

Teaching Experience

AθNLP 2024 / 2025 **Sep 2024 / Sep 2025**
Summer School Teaching Assistant, [*code*]

- Developed the lab materials for sequence modeling and multimodal models

Data Mining and Machine Learning **Sep 2022 - present**
Teaching Assistant

- Contributed to the infrastructure and the coursework of the module
- Supervised groups of postgraduate students on their semester project

Introduction to Natural Language Processing **Sep 2023 - present**
Teaching Assistant, [*code*]

- Contributed to the infrastructure and the coursework of the module
- Supervised groups of students on their semester project

Technical skills

Programming Languages	Python, Matlab, R, C, C++, C#, Java, PDDL, Bash, SML
Frameworks	PyTorch, Hugging Face, PyTorch Lightning, scikit-learn, openCV Albumentations, YOLO, Faiss, Milvus, Trax, Keras,
Development Tools	Weights & Biases, Gradio, AWS, Git, Slurm, Docker, Kubernetes, Jenkins

Honors & Awards

Evaluating Multimodal Language Models as Visual Assistants for Visually Impaired Users **Jul 2025**
Pioneer Center of AI Best Paper Award, ACL 2025 SAC Highlights

Exploring the Role of Patch Resolution for Vision Transformers **Mar 2022**
University of Edinburgh, 2nd place in the annual IBM Awards

Invited Talks

An Efficient Training Pipeline for Reasoning Graphical User Interface Agents **Nov 2025**
Alan Turing Institute

Embodied MultiModal Agent for Language-guided Action Execution in 3D Simulated Environments **Oct 2022**
CDT in Robotics and Autonomous System Conference, [*slides*]

Academic Service

ICLR Reviewer **Jan 2026**

ARR Reviewer **Jun 2024 / Oct 2024 / Jun 2025**

IEEE SLT218 Conference **Dec 2018**
Conference Volunteer